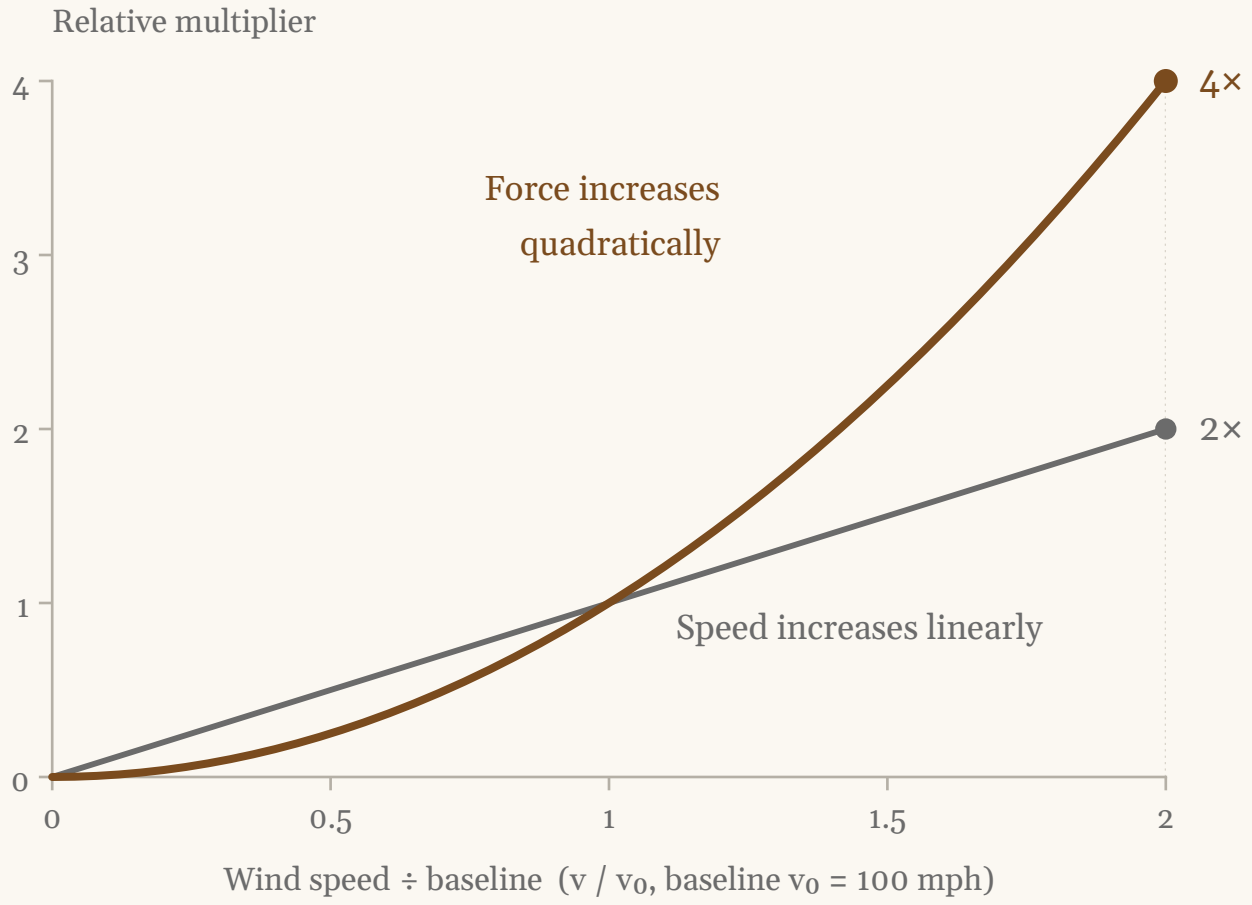
Twice the wind is four times the force



2× speed

4× force

100 mph → 200 mph

Air density, area, and drag coefficient held fixed.

Illustrative aerodynamic force, not measured hurricane damage.